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BIOLOGY
9610

BL02 – Biological systems and disease

Mark scheme

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1 8 1 X B L 0 2 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Marking guidance	Mark	Comments
01.1	Leu ;	1	
01.2	Mark as pairs: Substitution; A replaced by T;	2	 Accept: Insertion/Addition; T added after AA;
01.3	Change in shape / change in tertiary structure ;	1	
01.4	Cigarette smoke / ionising radiation – eg UV / X-rays / γ-rays / correct named chemical substance – eg formaldehyde / benzene / mustard gas ;	1	Accept tar
Total		**5**	

Question	Marking guidance	Mark	Comments
02.1	Xylem ;	1	
02.2	In Cutting 1 with xylem intact – film blackened up to top of cutting / ^{32}P moves up to top of cutting ; In Cutting 2 with xylem removed – film blackened up to (base of) ring / ^{32}P moves only up to (base of) ring ;	2	Must mention presence/absence of xylem at least once for full marks
02.3	In cutting 2 – slight blackening above ringed region / a little ^{32}P above ringed region ;	1	
02.4	Cutting 2 – black to top of cutting / sucrose carried to top of cutting ;	1	
Total		**5**	

Question	Marking guidance	Mark	Comments
03.1	Most bacteria have one circular chromosome ;	1	
03.2	Flagellum for locomotion ;	1	
03.3	Chloride ions leave epithelium cell/enter lumen of colon ; Water potential/solute potential lowered in lumen ; Water enters lumen (from cells) by osmosis/diffusion ;	3	Ignore Na^+ ions Allow sodium chloride
03.4	Whole toxin/subunit A would activate water loss from epithelium ;	1	Allow would cause symptoms of cholera / the disease
03.5	2.14 / 2.1 ;	1	
03.6	Quicker / no lag time / longer-lasting ;	1	
03.7	First dose: Small number of appropriate B cells present ; Role of antigen presenting cells ; B cells multiply → plasma cells → antibodies ; and → memory cells ;	3 max	
03.8	Second dose: Large number of memory cells → repeat in larger amounts ;	1	
Total		**12**	

Question	Marking guidance	Mark	Comments
04.1	A anaphase ; B telophase ; C prophase ;	3	
04.2	Spindle (fibre) ; Separation of chromatids ;	2	
04.3	Use of $\frac{\text{Number of cells in mitosis}}{\text{Total number of cells}} = \text{mitotic index} / \frac{3}{0.20}$; 15 ;	2	Correct answer = 2 marks (Ignore working)
04.4	Included only cells with visible nuclei or chromosomes ;	1	
04.5	Count (many) more cells ; Number counted by student not necessarily representative / may be variations across the tissue / small change in number of cells in mitosis → a large change in mitotic index ;	2	Allow exemplar figures – eg 4/15 → 0.27, 2/15 → 0.13
Total		**10**	

Question	Marking guidance	Mark	Comments
05.1	Correctly labelled on Figure X: Capsid ; Glycoprotein ;	2	
05.2	Integrase: Joins (double-stranded) DNA into host cell chromosome/into host DNA ; Reverse transcriptase: Makes (single-stranded) DNA complement of RNA ; DNA polymerase: Makes double-stranded (c)DNA from single-stranded DNA ;	3	
05.3	A mutation causes a change in the structure of a protein / enzyme / integrase / reverse transcriptase ; (Mutant) ernzyme is now effective in the presence of the (single) drug ; So virus can replicate and infect other (CD4 helper T-) cells ; With 2 drugs: unlikely to have >1 mutation in one virus ; So at least one enzyme is inhibited and virus cannot replicate ;	4 max	
Total		**9**	

Question	Marking guidance	Mark	Comments
06.1	Run at constant speed – so can compare / gives valid comparison ; Constant environment – otherwise changes in temperature / wind / rain would give invalid comparison ;	2	
06.2	Use of CO = HR x SV or CO = 148 x 38 ; Answer: 5.62 / 5.6 ;	2	Allow 2 marks for correct answer and ignore working Allow maximum 1 mark for 5624/5.624 and ignore working
06.3	Person 1's SV > Person 2's SV ; Person 1 has larger ventricle(s)/thicker (muscular) wall in ventricle(s) ;	2	 Allow left ventricle Ignore right ventricle
06.4	Atheroma formation/lipid deposits ; In arteries to heart wall/in coronary arteries ; Cigarette smoking → release of clotting factors ; Narrowing of arteries/(partial) blocking of arteries ; Lower blood flow to heart muscle/to walls of ventricles ;	4 max	Allow fatty deposits/cholesterol Allow reduced oxygen/glucose supply to heart muscle (eg re CO binding to haemoglobin) Allow lower respiration rate/energy release Do **not** allow reference to energy 'production'
Total		**10**	

Question	Marking guidance	Mark	Comments
07.1	Method: Cut from tree, put base in water & cut again ; Reason: Prevent air entering xylem / if air enters xylem stops water flow ; Method: Submerge apparatus in water for assembly ; Reason: To fill completely/to prevent air locks ; Method: Ensure leaves are kept dry/dry leaves with towel ; Reason: Water on surface blocks stomata/will reduce transpiration ; Method: Ensure base of stem (slightly) wider than plastic tube **or** seal joints with Vaseline ; Reason: Air-tight/prevent air entering ;	4	For full marks – 2 method points + reason for each Allow up to max. 3 for any 3 method points with no/incorrect reasons Allow up to max. 2 for any 2 reasons with no method
07.2	D	1	
07.3	On graph: Axes: correct orientation + appropriate scales + labelled inc. units ; Points plotted correctly ; Appropriate straight line of best fit;	3	

07.4	Answer: 52 ;	1	Allow ecf from candidate's graph
07.5	Calculation: πx $(0.5)^2$ x [slope from 07.4] x 60 ; Answer: 2.45 x 10^3 ;	2	Correct answer = 2 marks and ignore working Allow 1 mark for answer correct but wrong order of magnitude / not in standard form / not 3 sig figs Allow 2 marks for answer correct for candidate's graph / from answer to 07.4
07.6	Require water loss/evaporation from the plant only ;	1	Ignore it would not give an accurate answer
07.7	Transpiration/evaporation is the main cause of water entry/water movement ; Very little water remains in the plant/is used by the plant/nearly all water entering is transpired/evaporated ;	2	Do not allow <u>all</u> water taken in is lost
Total		**14**	

Question	Marking guidance	Mark	Comments
08.1	Stomach secretes protease / endopeptidase ; Endopeptidase action described–cutting in middle of polypeptide ; Hydrolysis/breaking peptide bonds ; Pancreas secretes more endopeptidases ; Pancreas secretes exopeptidases ; Exopeptidase action described – cutting off terminal amino acid ; Membrane-bound dipeptidases/dipeptidases within epithelial cells of small intestine ; Products – amino acids (+ small peptides) ;	5 max	Allow pepsin(ogen) Do not allow incorrectly named protease Allow trypsin(ogen)/chymotrypsin(ogen) Allow carboxypeptidase/aminopeptidase Allow enterokinase and activation of proenzymes
08.2	Facilitated diffusion of amino acids if high concentration in ileum ; Co-transport with Na^+ ions ; As Na^+ diffuses down its gradient ; Na^+ ions removed from epithelium into blood by active transport/described – eg pumped using energy from ATP ; Facilitated diffusion of amino acids into blood ;	5 max	Allow diffusion through channel protein Allow in exchange for K^+ ions/Na^+-K^+-dependent ATP-ase Allow additional points: • Facilitated diffusion of di-/tri-peptides into epithelial cells/co-transport with H^+ ions • Ref. to dipeptidases and tripeptidases in cytoplasm of epithelial cells
Total		**10**	